



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

Lakehouse Local Runtime Setup - First Use

Install only when C3-04-L03 begins; no Spark, Kafka or Airflow required for this phase.

Artifact	C3-049
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



1. Create a clean phase workspace

Use the normal Course-3 Python environment or create a phase-specific venv. Current PyArrow 25.0.1 supports Python 3.14, and current deltalake 1.6.3 provides Python 3.14-compatible wheels.

```
mkdir -p ~/course3/lakehouse/c3_04
cd ~/course3/lakehouse/c3_04
python3 --version
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
```

2. Install the pinned learner runtime

```
python -m pip install pyarrow==25.0.1 deltalake==1.6.3 "pytest>=9,<10"
python -m pip check
```

Expected: pip check reports no broken requirements. If install fails, save the real error; do not invent Parquet/Delta output.

3. Verify versions

```
python - <<'PY'
import pyarrow, deltalake
print('pyarrow', pyarrow.__version__)
print('deltalake', deltalake.__version__)
PY
```

Expected version line: pyarrow 25.0.1; deltalake 1.6.3.

4. Run the supplied checks

```
unzip C3_051_PRACTICE_STARTER_PROJECT_RC1.zip
cd practice_project
python scripts/00_runtime_check.py
python -m pytest -q
python scripts/static_project_audit.py
```

Resource-safe rule

Keep Airflow, Spark and Kafka closed for this phase. C3-04 deliberately teaches Parquet/Delta mechanics without requiring a heavy multi-service stack.